

POTPRESS · DESIGN PRECIS

PotPress design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried in: ISO 13857 desk check now as a hold point, pump slot inner shield, handle operating space, working force found in trials from about 2 t, demolding confirmed with potters
0.8	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.7	2026-09-27	Amish Chadha	Cost overrun decided by Amish on 2026-09-27; budget \$1,060, R10 met on paper
0.6	2026-09-26	Amish Chadha	Guarded version (decided by Amish 2026-09-26)
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish; budget \$990, R10 met on paper
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); bolted upright joints, hinged rail extension, pin-presence interlock, lead-free scrap, budget \$930; numbers from PPR-CAL-001 v0.2

0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices decided by Amish (PPR-DDR-001); numbers corrected to PPR-CAL-001 (shell molds, 2 x UPN 160 base, one 60 mm pin, 2 x 2 QC rack, cost)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, material flow, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

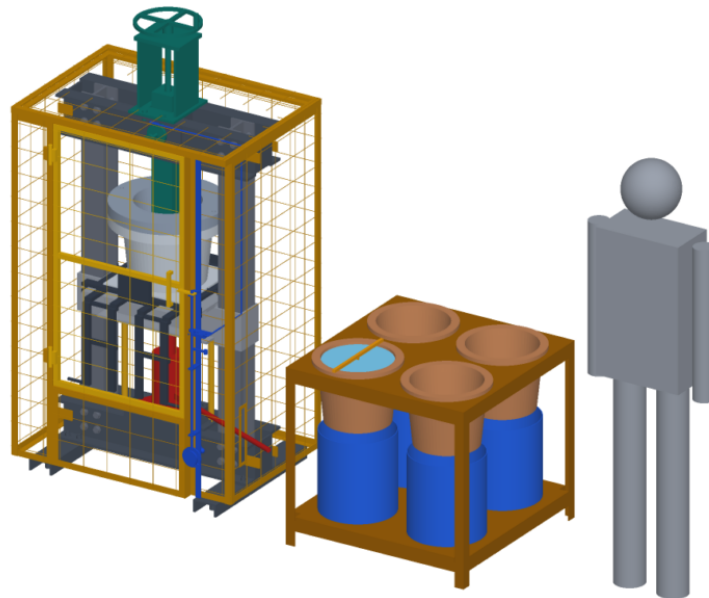
PotPress design precis

PotPress is a hand-pumped hydraulic press that forms one ceramic pot filter per stroke between a cast aluminum female mold and male mold, plus a four-station rack for the standard one-hour flow-rate test. A 20 t bottle jack on the base lifts a guided platen carrying the female mold against a fixed male mold, whose stem is held by one load pin in the top crossbeam for pressing and raised by a hand crank for loading. One parametric filter geometry drives the molds, the 3D-printed casting patterns and the printed flow gauge. The TRL 3 calculations (PPR-CAL-001) give a cycle of about 5.2 min and about 69 pots per 6 h, a strong frame at 1.5 times the jack rating, every part under 40 kg with the frame bolted at the upright joints, and, in the guarded version, a parts cost of \$1,051 against the \$1,060 value-engineering target (the target stood at \$990 before the guards were added, PPR-DDR-003). The press works inside fixed welded-mesh guards with a hinged front gate that is interlocked with the jack release, as decided by Amish on 2026-09-26 (PPR-DDR-003). The other design choices below were decided by Amish on 2026-09-25 (PPR-DDR-001 and PPR-DDR-002).

PotPress

github.com/BoujeeEnjinia1701/potpress

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey figure: 1.75 m person for scale

Figure 1. PotPress with the 2 x 2 QC flow-test rack and a 1.75 m person for scale. The press is shown closed at the end of a pressing stroke, inside its mesh guards with the front gate closed; the mesh is drawn at every eighth wire.

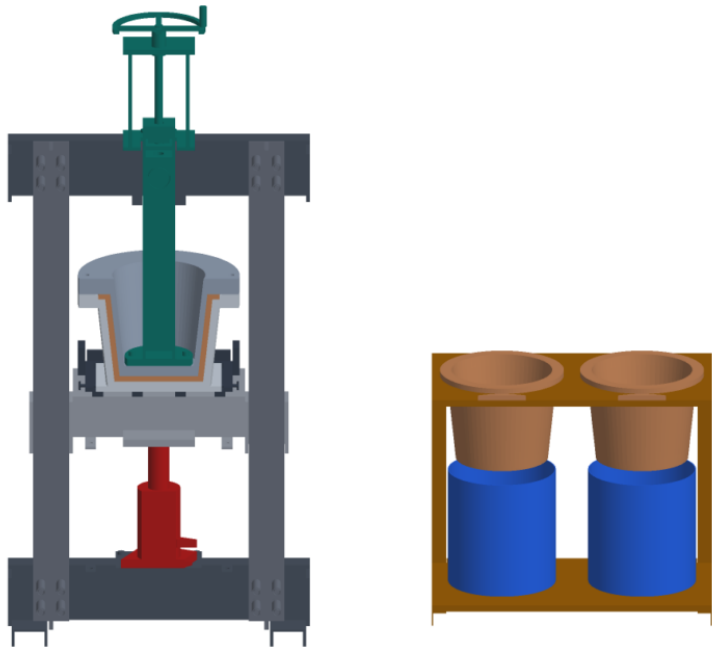
■ How it works

1. **Load.** With the platen down and the jack release open, the female mold rim is at about 921 mm. The operator opens the front gate, swings the hinged rail extension up onto its stop lugs, slides the carriage out 430 mm on the rails, lays a polyethylene release liner in the female mold, places a weighed charge of about 7.5 kg of clay, burnout and water mix, and slides the carriage back to its end stop.
2. **Lower the male mold.** A handwheel on top of the frame turns a self-locking Tr24 x 5 lead screw that runs the male mold stem down 200 mm (40 turns) through the top crossbeam until the pin bores line up. The operator inserts the 60 mm load pin through the beam and stem. From then on the pin, not the screw, carries the press force.
3. **Press.** The operator closes the front gate, which lets the guard-locking interlock free the jack release T-handle; closing the release locks the gate shut. The operator then pumps the jack through the slot in the right side guard, about 70 strokes. The platen, guided by sleeves on both uprights, lifts the female mold about 110 mm and squeezes the charge into the gap between the molds. The male flange centers in a tapered lip on the female mold and lands on a stop face, so the 15 mm wall and rim are set by the molds, not by how hard the operator pumps. Excess mix squeezes into a flash groove at the rim for trimming.
4. **Open and demold.** The operator opens the jack release with the T-handle outside the guard; the platen's weight and two springs return it, and only then does the interlock let the gate open. The operator pulls the pin and cranks the male mold up, leaving its tip 70 mm above the female rim. The carriage slides out toward the operator and tilts on its pivots so the pot, still in its liner, turns out onto a drying board. Between loads the rail extension folds down, so the press stands 840 x 655 mm.
5. **Dry, fire, treat.** Drying, firing and silver treatment follow the factory's existing practice and are outside PotPress.
6. **Test.** Fired pots are soaked, hung by their rims in the QC rack over collection buckets, filled to 10 mm below the rim and read after one hour with the printed T-gauge. The reading is corrected to 25 °C with the water temperature.

PotPress: cutaway

github.com/BoujeeEnjinia1701/potpress

CONCEPT, NOT FOR FABRICATION



Front half removed, guards and gate left out; seen from the front and above, 18 deg elevation

Figure 2. Cutaway through the closed press and the QC rack, showing the pressed pot between the molds, the shell molds and the load path.

PotPress: material flow per filter, mixed charge to passed filter (all values are estimates)

github.com/BoujeeEnjinia1701/potpress

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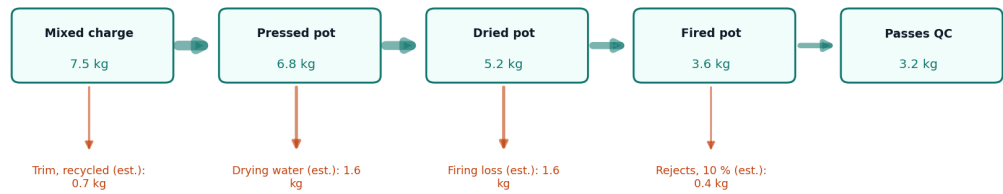


Figure 3. Material flow for one filter, from mixed charge to a filter that passes QC, in kg. All values are estimates from PPR-CAL-001.

■ Main components

Numbers match the exploded view (Figure 4), `bom/bom.csv` and the general arrangement drawing PPR-DWG-001.

Table 1. Main components.

#	Component	Choice	Notes
1	Base beam and feet	Two UPN 160 channels 840 mm with a 100 mm gap, 20 mm jack plate, two UPN 100 feet bolted on	Same section as the top beam; the TRL 2 single UPN 100 would reach about 1,071 MPa
2	Uprights	Two UPN 100 channels back to back each side, about 1.40 m, sandwiched in the beam gaps at 600 mm centers and bolted to both beams with 4 x M20 8.8 per joint	54 MPa in tension, bolts 75 MPa in double shear at 294 kN; 29.4 kg per pair
3	Top crossbeam	Two UPN 160 channels, 12 mm web doublers and a 62 mm pin bore at the center, UHMW-PE stem guides	190 MPa at 294 kN
4	20 t bottle jack	Standard jack, 150 mm stroke, about 60 mm ram	Uses 110 mm of stroke
5	Moving platen	Two UPN 140 channels, 6 mm deck, jack pad, 150 mm guide sleeves on the uprights	38.9 kg, the heaviest part as fabricated
6	Return springs	Two tension springs	Gravity returns the platen; the springs keep it seated
7	Rails, hinged extension and carriage	Four flat-bar rails at 60 and 170 mm each side, fixed to 320 mm in front of the axis, with a 330 mm extension on barrel hinges and stop lugs that folds down; carriage frame with handle and tilt pivots	Moves the female mold out of the pinch zone; 43 N·m on the hinge with the carriage out
8	Female mold	Cast aluminum shell, 20 mm behind the cavity, 30 mm solid base, 410 mm flange with a locating lip, rim counterbore and flash groove; 24.3 kg	A solid block would weigh about 57 kg
9	Male mold	Cast aluminum hollow plug, 15 mm shell and 30 mm tip, 390 mm rim-forming flange; 17.9 kg	A solid plug would weigh about 41 kg
10	Male mold slide	SHS 90 x 8 stem with a pin block, one 60 mm 42CrMo4 load pin, crank bracket, Tr24 x 5 lead screw and 260 mm handwheel	The screw only lifts; the pin carries the load
11	Pressed filter pot	The product, shown for context	Not costed
12	QC flow-test rack	2 x 2 stations, 780 x 780 x 550 mm, steel angle and plywood	Within R11's 0.8 x 0.8 m
13	Printed T-gauge	PETG dipstick; 1.0 L is 16.7 mm on the scale	Scale generated from the geometry
14	Collection buckets	20 L food-grade HDPE	
16	Fixed mesh guards	Welded mesh 12.7 x 12.7 x 1.6 mm (about 11 mm clear) on 25 x 25 x 3 angle frames: both sides, back, two front strips, a lower front panel and a roof around the top beam; standoffs to the beams; a framed pump handle slot with a brush strip in the right side	Mesh planes 940 x 675 mm; about 4.3 m ² of mesh
20	Front gate	532 x 1,063 mm, 20 x 20 x 2 tube frame with the same mesh, two lift-off hinges on the left front post, pull handle and striker tongue; opens outward to the left	Clears the carriage, molds and rail extension (540 mm opening)
21			No electrics

#	Component	Choice	Notes
	Gate interlock and release extension	Mechanical guard-locking bolt on the right front post, link rod and lock bar to a blocking cam on the jack release extension; T-handle outside the lower front panel; the pin-presence plunger acts on the same lock bar	

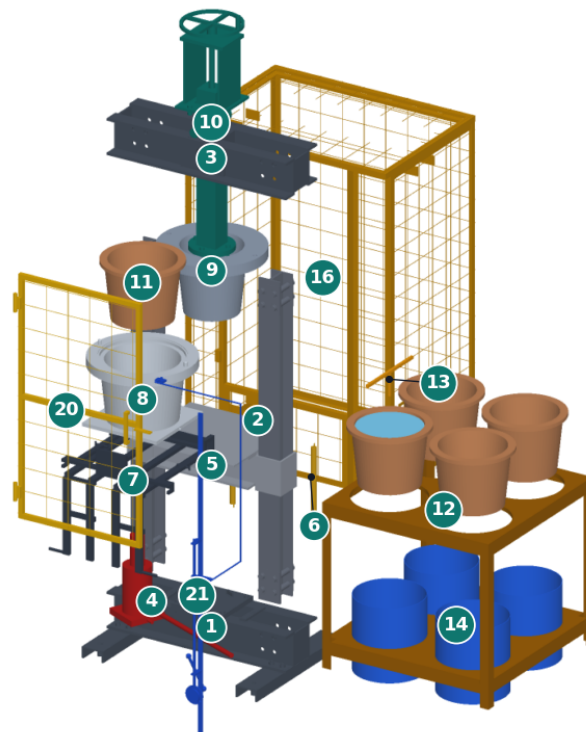
Casting patterns, release liners, fasteners and consumables, and the QC timer and thermometer are BOM items 15 and 17 to 19 and are not modeled. The guards are modeled with their mesh drawn at every eighth wire (101.6 mm) so the views stay readable; the appearance model draws every wire.

PotPress: exploded view

github.com/BoujeeEnjinia1701/potpress

CONCEPT, NOT FOR FABRICATION

- 1 Base beam, feet and jack plate
- 2 Uprights with bolted joints
- 3 Top crossbeam with pin doublers
- 4 20 t bottle jack
- 5 Moving platen with guide sleeves
- 6 Return springs
- 7 Rails, folding extension and carriage
- 8 Female mold (cast aluminum on a steel base plate)
- 9 Male mold (cast aluminum)
- 10 Male mold slide, load pin, crank
- 11 Pressed filter pot
- 12 QC flow-test rack (2 x 2)
- 13 Printed T-gauge
- 14 Collection buckets
- 16 Fixed mesh guards (sides, back, front strips, roof)
- 20 Front gate, mesh in a tube frame, with hinges and tongue
- 21 Gate and pin interlock, release shaft



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 4. Exploded view with numbered callouts matching the BOM.

■ Key numbers

All values are estimates from PPR-CAL-001, which lists its assumptions and the results against every requirement.

Table 2. Key numbers at TRL 3.

Quantity	Value	Basis
Filter volume	12.30 L to the brim; 9.91 L working	Inner rim 280 mm, floor 230 mm, depth 240 mm
Pot masses	Charge 7.47 kg; pressed 6.79 kg; fired 3.61 kg	RDI-C recipe, 1.64 kg/L pressed
Mean pressure	0.52 to 1.05 MPa at 5 to 10 t; 2.10 MPa at 20 t	0.0935 m ² projected
Frame at 294 kN	Beams 190 MPa, uprights 54 MPa, pin 258 MPa (yield 650), joint bolts 75 MPa	S275; 42CrMo4 pin; M20 8.8 bolts
Deflection between molds	0.61 mm at 10 t; 1.84 mm at 294 kN	R3 (1 mm at 10 t) met on paper
Travel	110 mm pressing of 150 mm stroke; 310 mm opening against 270 mm needed	R5 met
Cycle	5.2 min; 69 pots per 6 h	R4 met
Wall evenness	±1.40 mm as cast; ±0.43 mm finished to templates	R2 at risk
Flow gauge	0.1 L is 1.69 mm; 2.3 % per °C near 25 °C	R6 met
Size and mass	Press 840 x 655 x 1,806 mm unguarded; 940 x 700 mm guarded, with the rail extension folded; 298 kg plus 7 kg of joint bolts and about 45 kg of guards (estimate); heaviest part 38.9 kg; rack 780 x 780 mm	R8 met on paper; R11 met on paper at the 0.7 m limit
Guarding	Mesh 12.7 mm pitch; nearest moving parts behind the mesh: about 105 mm at the back (platen deck), 109 mm at the sides (platen sleeves), 110 mm at the front (carriage handle)	Mesh distances assumed adequate; the ISO 13857 desk check is done now, signed by a competent person, and is a hold point before any force above hand pressure (decided by Amish, 2026-10-02); the 30 mm pump slot needs a fixed inner shield behind it (decided, 2026-10-02)
Cost	Press \$970, QC rack \$81, total \$1,051	within the \$1,060 value-engineering target, \$9 (0.8 %) under

The working force is still an assumption. Henry, Maley and Mehta (2013) formed round-bottom filters with a 2 t car jack ([IJSLE 8 \(1\)](#)), which suggests the real force may be well below 5 to 10 t; the frame is sized for the full 20 t jack either way. Decided by Amish on 2026-10-02: every structural check stays at the full 20 t jack, and the working force is a process setting found in pressing trials with a partner factory, starting near 2 t and stepping up.

■ Key design choices

These were decided by Amish on 2026-09-25: go with recommendation (PPR-DDR-001 items 1 to 8 and 10 to 15; PPR-DDR-002), except the guarded version, decided by Amish on 2026-09-26 (PPR-DDR-003).

- **Jack below, pushing the female mold up.** A bottle jack must stand upright, so it sits on the base and lifts the platen, and the male mold hangs from the top beam.
- **Male mold lift by hand crank with one load pin and an interlock.** A standard jack gives only 150 mm of stroke, so the crank provides the other 200 mm of opening. The TRL 2 pair of 30 mm pins fails in bending by calculation;

one 60 mm alloy-steel pin carries the force, and a mechanical pin-presence interlock stops the jack release valve closing unless the pin is fully home.

- **Molds cast in aluminum from printed patterns, from lead-free scrap.** Cast as shells to keep them under 25 kg, with the cavity finished by hand to printed templates. Scrap must be lead-free (for example cast wheels or pistons; no free-machining bar). The fiber-reinforced concrete backed variant stays documented as a low-cost alternative; building it is TRL 4 work and on hold.
- **Welded subassemblies, bolted at the upright joints.** The base beam, top beam, uprights and platen are welded separately and joined with 4 x M20 8.8 bolts at each of the four upright joints, so no part exceeds 40 kg. Fully welded, the frame would be one 147 kg piece.
- **Molds close on a metal stop.** The male flange lands on the female stop face, so frame stretch does not set the wall; R3 judges deflection at the 10 t working force.
- **Demold by sliding out and tilting the carriage,** on a rail extension that folds down between loads.
- **Manual QC rack with printed T-gauge.** The rack is 2 x 2 stations in 0.8 x 0.8 m; a load-cell logger stays a later option.
- **Default acceptance band 1.0 to 2.5 L/h in the first hour, corrected to 25 °C.** Each factory may set its own band.
- **Value-engineering target \$1,060.** This is a hypothetical control target, not a limit. The priced BOM was \$984 after the bolted joints and rail hinges, against a \$990 target. The guarded version adds \$67 and brings the BOM to \$1,051, \$61 over that earlier target; the target was set at \$1,060 on 2026-09-27, since guarding is a safety requirement (PPR-DDR-003). The estimate is within the target, \$9 (0.8 %) under.
- **Guarded version (decided by Amish, 2026-09-26; PPR-DDR-003).** Fixed welded-mesh guards on the sides, back and roof, and a hinged front gate with a mechanical guard-locking interlock on the jack release. A hand pump has no power to switch off, so the interlock acts on the one thing that lets the jack build pressure: the release valve. With the gate open the release cannot be closed, so pumping moves nothing; with the release closed the gate cannot be opened. This was chosen over two-hand control, which does not suit a one-handed pump lever, and over hold-to-run alone, which a hand pump already is but which leaves the other hand free to reach the molds.

■ Safety

Safety: PotPress is a 20 t hydraulic press with heavy moving parts, and the workshop around it handles silica-bearing clay dust and silver compounds. Treat each of these as a hazard at every stage.

- **Crushing and pinch points.** The gap between the molds, the platen and the uprights, and the carriage rails can crush fingers and hands. The guarded version (PPR-DDR-003) encloses them: fixed welded-mesh guards on the sides, back and roof, fixed front strips and a lower front panel, and a hinged front gate. The gate's guard-locking interlock holds the jack release open unless the gate is shut, and holds the gate shut while the release is closed. The pump handle works through a framed slot in the right side guard, well below the platen, and the release T-handle sits outside the lower front panel. The carriage has an end stop so it cannot be pushed past the male mold. Keep fingers clear of the rail extension hinges when folding it, and deploy it only onto its stop lugs; the extension cannot be deployed with the gate shut.
- **Guard openings (stated assumption).** The mesh is 12.7 mm (1/2 in) square welded mesh with 1.6 mm wire, about 11 mm clear, and the nearest moving parts are about 105 to 110 mm behind the mesh. ISO 13857 (safety distances to prevent hazard zones being reached by upper and lower limbs) is the reference; this precisely assumes, without having checked its tables, that these openings and distances keep fingers out of the pinch

zone. The reach over the roof is also unchecked. Decided by Amish on 2026-10-02: the ISO 13857 check is done now as a desk check from the tables and the model's distances, signed by a competent person, and is a hold point before any force above hand pressure. The pump handle slot, 30 mm wide since the design for construction, is approved only with a fixed inner shield or tunnel behind it that keeps the platen and molds out of arm's reach, sized in that check; a brush strip does not count as a guard. If the standard asks for more, use a finer mesh or move the panels out. The 158 mm the pump handle stands outside the guard while in use is operating space, marked on the floor layout. Never remove a guard or defeat the interlock, and inspect both before each shift.

- **Load pin and stored energy.** One 60 mm pin carries the whole press force. Never press unless the pin is fully home through both beam webs and the stem; the guard includes a pin-presence interlock that stops the jack release closing until the pin is home. Check the bolted upright joints for tightness before each shift. Never exceed the jack rating, never add a cheater bar to the pump handle, and never adjust the jack's relief valve. Release pressure fully before opening the gate or pulling the pin.
- **Falling and tipping.** The molds weigh about 24 and 18 kg and the press about 305 kg with its bolts, with its center of mass about 0.8 m up. A push of about 750 N at 1 m tips it forward, so anchor the base to the floor, lift molds with two people and wear safety boots. The lead screw is self-locking, so the male mold holds its height if the handwheel is released; still keep hands out from under it.
- **Silica dust.** Dry clay and rice husk ash contain crystalline silica, which causes silicosis. Keep mixing and trimming wet, sweep damp, and wear a fitted P2 or N95 respirator for dry work.
- **Molten aluminum and alloy.** Casting is done by a foundry, not at the press. Scrap containing lead (free-machining alloys) must not be used for mold faces that touch a drinking-water product.
- **Silver compounds.** Silver nitrate is corrosive to skin and eyes and toxic to aquatic life; handle colloidal silver with gloves and eye protection and never pour it into waterways. Silver treatment is outside PotPress but happens in the same workshop.
- **Heat.** Kilns reach about 900 °C. Kilns are outside PotPress but share the site.
- **Water safety claims.** PotPress makes consistent pots; it does not certify them. A filter that passes the flow test is not proven to remove pathogens. Factories must verify bacteria removal by laboratory testing and follow the CMWG recommendations.

■ Open questions

- What pressing force gives a well-consolidated wall with a typical mix? Found in pressing trials with a partner factory, starting near 2 t and stepping up (decided, 2026-10-02).
- Can hand finishing to printed templates reach ± 0.3 mm on a 400 mm casting (R2, R7)?
- Handwheel height of about 1.8 m: a side crank through a bevel gear would be easier to reach.
- Confirm the demolding method with the potters at the first partner factory; the tipping pins stay for the prototype, and if the potters object the pot is lifted out in its liner (decided, 2026-10-02). Check whether a liner leaves marks that affect flow.
- Choose the first partner factory or organization (left open under the portfolio rule).

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: PPR-DWG-001.